Milano-Cortina 2026 Winter Olympics: Behind the Scenes
VizLaLune

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`To the GitHub Repository!`

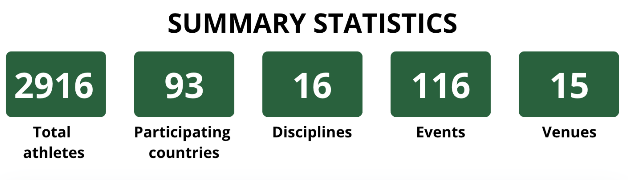
This project provides an interactive and visual exploration of the Milano-Cortina 2026 Winter Olympics. Rather than focusing on results or rankings, we explore the structure of participation itself: how many athletes compete, which countries do they represent, which sports they compete in, whether athletes primarily compete in individual, dual, or team events, and whether gender representation is balanced across disciplines. Unlike most Olympic visualizations which focus on medal counts, our approach centers on participation structure and uses novel animated storytelling, including an athlete travel animation and a moving bubble conclusion, to make the data exploration engaging rather than purely analytical. The target audience includes sports fans, data journalists, and anyone interested in global representation in international competitions.

Part 1 — Global Overview (Core)

Questions: How many athletes participate? How many countries? Which sports are most represented? How are the Games split among its venues?

This section is the entry point, following the Visual Information Seeking Mantra (Shneiderman, 1996): overview first, zoom and filter, details on demand.

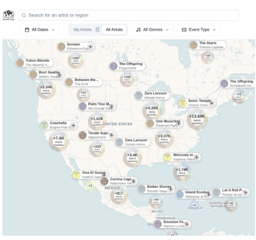
Fig 1.1 — Key statistics (bubbles)



Total athletes, countries, disciplines, events, and venues as animated bubbles. When a country is selected elsewhere, the bubbles update to reflect that country's participation only.

Lectures: W4.2, W6.1, W6.2, W7.2, W11

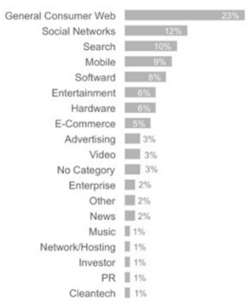
Fig 1.2 — Interactive venue map



Map of Italian host venues. Hover over a venue to expand it and view the disciplines taking place, with clickable links to the official venue/discipline pages. Can be displayed side by side with Fig 1.3.

Lectures: W4.2, W5.1, W6.2, W7.2, W8.2

Fig 1.3 — Venue exploitation (bar chart)



Sorted horizontal bar chart showing the percentage of event load per venue. Linked with Fig 1.2.

Lectures: W4.2, W5.2, W6.2, W7.2, W11

Fig 1.4 — Treemap (disciplines → events)

Hierarchical treemap splitting first into disciplines, then into individual events. Area is proportional to medal count, giving insight into which sports contribute most to the overall medal tally. Hovering reveals the winners.

Lectures: W4.1, W4.2, W6.2, W10

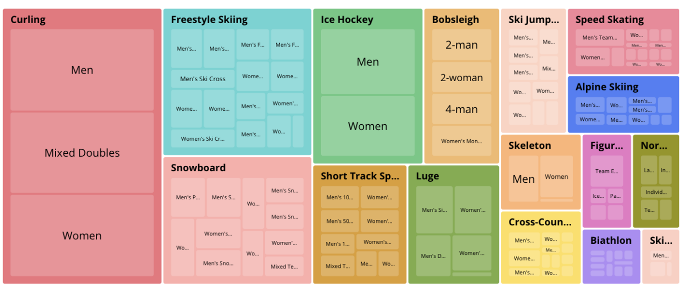
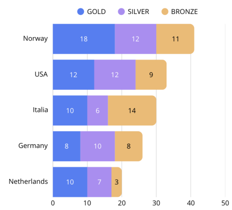


Fig 1.5 — Medal race (horizontal bar chart)



Classic medal count visualization with filters for discipline, event, country region, gender, and event type (individual / duo / team).

Lectures: W4.2, W5.1, W5.2, W6.2, W11

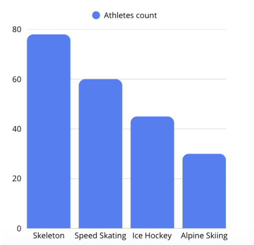
Data: `athletes.csv`, `medals.csv`, `medallists.csv`, `venues.csv`, `schedules.csv`

Tools: D3.js for data loading, bubble layouts, bar charts, treemaps; Leaflet for the interactive map.

Part 2 — Geographic Structure (Core)

Questions: Which countries participate the most? Which sports are dominated by which nations?

Fig 2.1 — Athletes per discipline (bar chart)



Global athlete count per discipline. When a country is selected on the map, updates to show only that country's discipline breakdown.

Lectures: W4.2, W5.1, W5.2, W6.2, W11

Fig 2.2 — Athlete travel animation (world map)

One dot per athlete travels from their country of origin to the host venues. The map can be filtered by clicking a country or selecting a global region. Hovering shows key statistics including athlete count and gender split.

Lectures: W4.1, W4.2, W5.1, W5.2, W7.2, W8.2

Data: `athletes.csv` — grouped by `country_code`, `discipline`, `gender`

Tools: D3.js, TopoJSON (world)

Part 3 — Gender & Team Representation (Core)

Questions: Are sports gender-balanced? Which disciplines are dominated by team vs. individual events?

A diverging horizontal bar chart with a central axis makes imbalances immediately visible. The chart reacts to the global selection state: when a country is selected on the map, it shows the gender split for that country specifically, revealing country-level patterns that would be invisible in the global view.

Fig 3.1 & 3.2 — Diverging bar charts

Fig 3.1: Men (left) vs women (right) per discipline. Filterable by region or country. Reacts to global selection state. Sortable by total athlete count or degree of imbalance with smooth D3 transitions.

Fig 3.2: Same structure: individual vs team athletes per discipline and country. A sync button links the filters of both charts simultaneously.

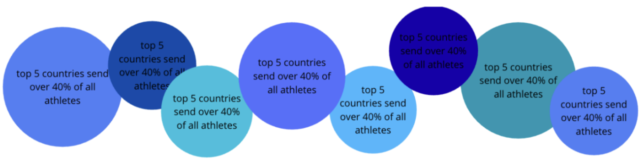
Lectures: W4.2, W5.1, W5.2, W6.1, W6.2, W11

Data: `athletes.csv` — grouped by `discipline`, `gender`, `event_type`
Tools: D3.js

Conclusion

The visualization tells a data story that progresses from the global (how many?) to the geographic (who?) to the thematic (how balanced?), following the structure advocated by Segel & Heer (2010) for narrative visualization.

Fig 4.1 — Moving bubbles



Animated bubble layout summarizing the main takeaways from the three sections: key patterns in participation, geography, and gender representation. Each bubble carries an insight label and animates into view as the user reaches the end of the page.

Lectures: W4.1, W4.2, W5.2, W6.2, W7.2, W11, W12.1

Technical Architecture

All sections share a single global selection state:

```
const state = {
  selectedCountry: null,
  selectedDiscipline: null
}
```

Any interaction updates this state and triggers reactive updates simultaneously across all views (linked views pattern, Week 5.1). This ensures the visualizations behave as a unified exploratory tool rather than three independent charts. All visualizations use D3.js (v7) with TopoJSON for world map and Leaflet for venue view. The site is a static HTML/CSS/JS page on GitHub Pages. Data is served as preprocessed CSV files.

Section	Tools	Data
Global overview	D3.js, Leaflet	athletes, medals, venues, schedules, medallists
Geographic map	D3.js, TopoJSON	athletes.csv
Gender & team	D3.js	athletes.csv

Breakdown — Core vs. Extra

Core — required for final delivery:

- Key statistics bubbles, venue map, treemap, and medal race (Part 1)
- Athlete travel animation with country filter and linked bar chart (Part 2)
- Gender diverging chart and solo vs. team chart with synchronized filters (Part 3)
- Animated conclusion bubbles
- Fully linked views via shared selection state

Extra — to be added if time allows:

- Comparison with past Winter Olympics (requires additional dataset)
- Visitor, economic, or environmental impact (requires additional dataset)

Prototype

The current prototype consists of a static HTML page with a sliding sidebar, the four-section layout, and SVG placeholder containers for each figure.

`github.com/com-480-data-visualization/VizLaLune`